Bridging ESG Text Analysis and Global Value Chain Resilience Through AI-Driven Innovation

Anonymous Authors

Abstract

Environmental, Social, and Governance (ESG) factors have become pivotal in shaping corporate sustainability and value creation across global supply chains. As ESG-related textual data continues to grow, the ability to accurately and efficiently annotate such information has emerged as a critical need for both investors and regulatory bodies. Traditional methods, however, often fall short in addressing the complexity and contextual nature of ESG texts, particularly in multinational environments. This paper introduces En-BERT, an innovative framework grounded in ensemble learning that integrates multiple pre-trained language models to enhance the accuracy, stability, and efficiency of automated ESG text annotation. We meticulously constructed a high-quality financial corpus by selecting seven companies with exemplary ESG performance across diverse industries. Through rigorous data selection, collection, and manual annotation protocols, we ensured the dataset's quality, representativeness, and comprehensive coverage of ESG practices. This foundation supports the robust training and superior performance of the En-BERT model. By open-sourcing both the dataset and the En-BERT model on GitHub, this research bridges innovations in natural language processing (NLP) with advancements in global value chain management. It offers actionable pathways for transforming textual ESG data into strategic assets that drive sustainable value co-creation. Our findings highlight how AI-powered text annotation can significantly enhance supply chain resilience, addressing a critical challenge for multinational corporations as they navigate geopolitical uncertainties and climate disruptions. Through precise and efficient ESG text annotation, En-BERT empowers enterprises to convert unstructured textual data into informed decisions, fostering transparency, sustainability, and competitive advantage in today's complex global business landscape.

Keywords: Financial Text Annotation; ESG Analysis; Green Investment; Sustainable Development; Supply Chain Resilience

1 Introduction

As the global influence of ESG (Environmental, Social, and Governance) factors continues to grow, their role in shaping multinational corporations' value creation across global supply chains has become pivotal for sustainable development [1, 2, 3, 4]. By 2018, over half of global asset owners had already incorporated ESG considerations into their investment strategies or were in the process of doing so [5, 6, 7]. In recent years, both investors and regulators have increasingly recognized ESG factors as critical indicators of corporate sustainability, with performance in environmental protection, social responsibility, and governance not only reflecting a company's intrinsic value but also forecasting its long-term development potential and risk profile [8, 9]. The year 2023 witnessed a watershed moment in ESG compliance when Tesla's German Gigafactory faced a $2.1B supply chain disruption due to inconsistent ESG disclosures between its Berlin facility and American headquarters [10]. This incident, stemming from mismatched environmental compliance reports and conflicting labor practice documentation, epitomizes the growing challenge of maintaining ESG coherence in global value chains –

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(GVCs). As multinational corporations expand across jurisdictions, the semantic heterogeneity in sustainability disclosures has emerged as a critical risk factor - one that conventional ESG assessment frameworks are ill-equipped to address.

By establishing a robust ESG governance framework and integrating targeted strategies across the environmental, social, and governance dimensions, firms can significantly mitigate adverse effects, potentially transforming controversies into opportunities for growth and reputation enhancement [10]. Recent studies by Dai and Tang [1] demonstrate that integrated ESG frameworks significantly strengthen supply chain resilience in post-pandemic global operations. This positions ESG factors as both risk sources and catalysts for value creation across geographically dispersed operations [11]. Moreover, the advent of advanced technologies has empowered global ESG governance, offering enterprises more efficient tools to address the complexities of worldwide operations.

Unlike conventional financial metrics or corporate social responsibility (CSR) evaluation frameworks, ESG assessment offers a systematic approach to measuring corporate sustainability across three interrelated non-financial dimensions: environmental, social, and corporate governance. Notably, the interplay between corporate performance and ESG-related controversies—such as those revealed through media reports and corporate disclosures—has emerged as a prominent research focus. For example, Elamer et al. [12] report a significant negative correlation between ESG controversies and corporate performance. Yet, establishing a robust ESG governance framework and integrating targeted strategies across the environmental, social, and governance dimensions can significantly mitigate these adverse effects, potentially transforming controversies into opportunities for growth and reputation enhancement[10],thereby positioning ESG factors as both risk sources and catalysts for value creation.

Despite the increasing utilization of ESG-related textual data in accounting and finance, its analysis remains a challenging task. The current approaches are often labor-intensive, subjective, and costly [13], with particular difficulties arising in processing news opinions and disclosure information where both efficiency and accuracy are paramount [14]. Moreover, the highly contextual nature of ESG content further complicates automated classification efforts[15]. Consequently, efficiently and accurately extracting ESG-related information from vast amounts of unstructured financial text has become a critical challenge. Traditional manual annotation methods, while typically accurate, struggle with ensuring consistency and objectivity . In response, the advent of Large Language Models (LLMs) has introduced promising new methodologies; however, high annotation costs and domain-specific limitations continue to hinder their broader application [16]. In contrast, pre-trained language models such as BERT have achieved impressive results in various natural language processing tasks, offering innovative approaches for automated text annotation [17, 18]. Nevertheless, these models still exhibit shortcomings in fully grasping the nuances of ESG concepts and in maintaining high annotation accuracy when applied to complex, specialized financial texts in multinational contexts, often resulting in deviations from expert standards [19].

To address these challenges, this paper introduces En-BERT (Ensemble Bert of data and model disturbance), an integrated BERT annotation model grounded in ensemble learning. By leveraging the complementary strengths of multiple pre-trained models, En-BERT aims to enhance the accuracy and stability of automated text annotation, particularly in complex, specialized financial texts across multinational contexts. Specifically, our approach comprises a rigorous data selection and annotation process. Seven companies with exemplary Refinitiv ESG ratings were carefully chosen through expert evaluation and industry screening. Their news texts, collected over a one-year period, were meticulously curated and manually annotated to construct a high-quality financial corpus. Building on this robust dataset, we designed a multi-model labeling system—En-BERT—that performs efficient, automated labeling across environmental, social, and governance dimensions. The proposed En-BERT addresses these challenges by enabling standardized ESG benchmarking across borders, which is critical for aligning corporate practices with global sustainability standards [20]. Experimental results confirm that En-BERT significantly reduces annotation errors and improves labeling accuracy while substantially lowering labor, time, and resource costs. The research framework of this paper is shown in Figure 1.

The contributions of this paper are summarized as follows:

(1) High-Quality Dataset Construction: We have developed a comprehensive financial text corpus annotated with E, S, and G tags. This dataset not only underpins our model training but also provides a valuable resource for subsequent research in ESG evaluation.

(2) Innovative Annotation Model: The proposed En-BERT model integrates multi-source pre-

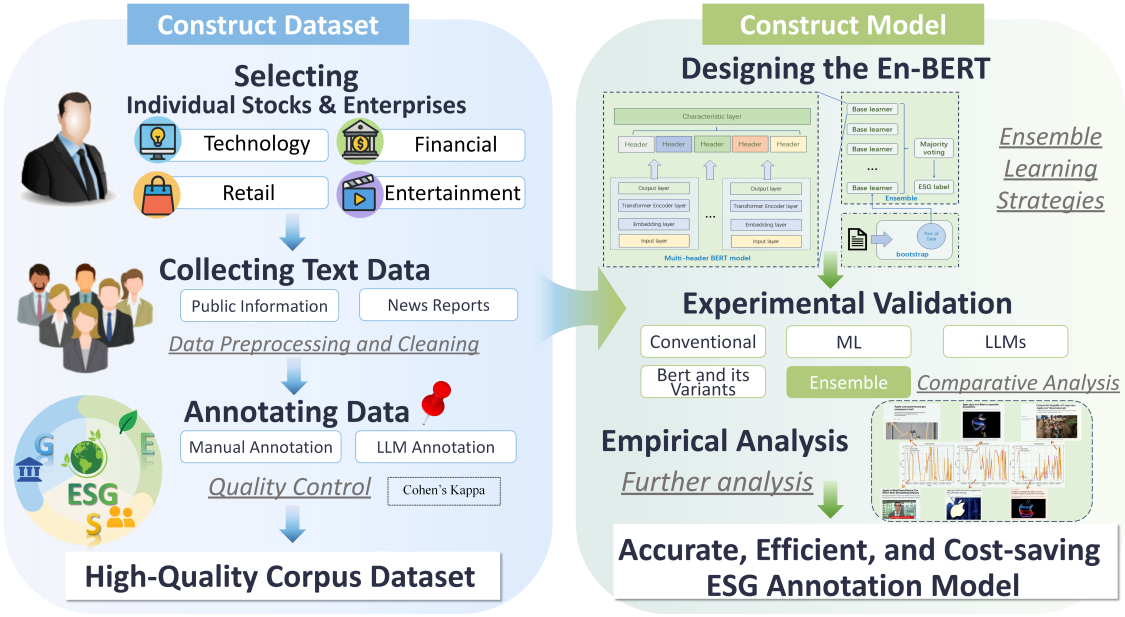


Figure 1: En-BERT: A Comprehensive Framework for ESG Text Annotation and Global Value Chain Applications

trained models to capture cross-cultural ESG expressions, providing a unified framework for global supply chain participants to assess sustainability performance. Compared to conventional approaches, it not only ensures and improves annotation accuracy but also significantly enhances labeling efficiency while reducing costs. This model offers a novel solution for ESG text annotation, laying a solid data foundation for more in-depth analyses in the field of green investment and supervision.

3) Open-Source Sharing and Technological Advancement: Effective ESG text annotation serves as the foundation for managing risks in GVCs. As shown in [11], inconsistencies in sustainability reporting across suppliers can lead to systemic vulnerabilities—a challenge En-BERT directly addresses through its high-precision automated labeling. We have already open-sourced both the dataset and the En-BERT model on GitHub, along with comprehensive test cases, example applications, API documentation, and a project maintenance plan , thereby offering new technological pathways for harmonizing ESG disclosures in fragmented global markets, particularly in emerging economies. This initiative not only enhances information disclosure and regulatory oversight but also empowers investment decision-making in financial markets. By providing a unified framework for ESG assessment, our open-source approach contributes to a more robust, inclusive, and sustainable financial ecosystem, bridging gaps in ESG practices across diverse economic environments.

2 Methodology

Our research proposes a novel framework for multi-model fusion in text classification, leveraging the semantic representation abilities of various pre-trained models. The framework integrates a hierarchical voting mechanism and multi-granularity feature fusion to enhance classification performance. The entire process is divided into four key stages: text preprocessing, semantic embedding extraction, decision fusion, and ensemble learning.Specifically, the proposed technical workflow presented by the research is shown in Figure 2.

Our theoretical framework is grounded in a probabilistic graphical model that governs the fusion process:

$$P(y|T,\mathcal{M}) = \sum_{z\in\mathcal{Z}} \underbrace{P(y|z)}_{\text{Decision Layer}} \cdot \underbrace{P(z|\phi(T),\mathcal{M})}_{\text{Semantic Encoder}} \tag{1}$$

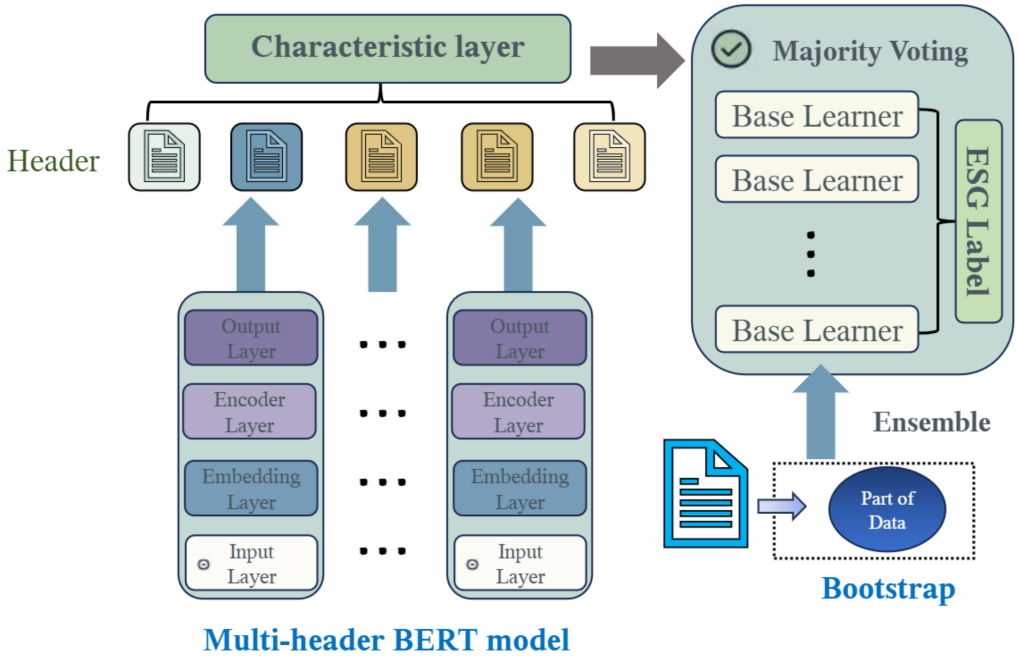


Figure 2: An innovative framework for the Optimized technical workflow incorporating Hierarchical Voting and Multi-granularity feature fusion

where $\mathcal{M} = \{M_i\}_{i=1}^{7}$ denotes the ensemble of pre-trained language models, each following architectural constraints $\mathcal{A}_i \subset \mathbb{R}^{d_i}$. The latent decision state $z \in \mathcal{Z}$ captures model agreement levels, and $\phi(T)$ represents our multi-granularity feature mapping function.

2.1 Text Preprocessing

The text preprocessing phase employs a heterogeneous subword segmentation strategy to process the raw input sequence. For each pre-trained model (including BERT, RoBERTa, and five other architectures), its tokenizer converts the raw text $T = \{w_1, w_2, \ldots, w_N\}$ into a specific subword sequence. The processed token sequence is generated through the formula:

$$T_{\text{proc}}(i) = [[\text{CLS}], s_1(i), \ldots, s_{M_i}(i), [\text{SEP}]] \tag{2}$$

where the number of subwords M_i is determined by the dynamic calculation:

$$M_i = \left\lfloor L_{\max} \times \frac{N_{\text{avg_len}}}{\text{len}} \right\rfloor \tag{3}$$

Here, $L_{\max}$ is the maximum sequence length, and $N_{\text{avg_len}}$ is the average token length across the dataset. The input tensor construction process is achieved through the tuple $X_{\text{input}}(i) = (I(i), M(i))$, where the index matrix $I(i)[j] = \text{Index}_i(s_j(i))$ performs vocabulary mapping, and the mask matrix $M(i)[j] = \mathbb{I}(s_j(i) \neq [\text{PAD}])$ identifies valid token positions using an indicator function. This hybrid strategy ensures optimal tokenization for each model while maintaining computational efficiency.

2.2 Semantic Embedding Extraction

The semantic embedding extraction phase utilizes a multi-level encoding mechanism to generate robust semantic representations. The initial embedding layer produces the semantic representation $H_i^{(0)} = E_i \cdot I(i) + P_i$ through a linear transformation, where the word embedding matrix $E_i \in \mathbb{R}^{|V_i| \times d}$ and position encoding $P_i \in \mathbb{R}^{M_i \times d}$ are adapted to each model's architecture. For standard BERT-like

models [21, 22], the semantic vector is extracted as:

$$h_i = H_i^{(L)}[0,:] \tag{4}$$

directly capturing the global semantics of the [CLS] token. For multi-label optimized models like DistilBERT [23] and RoBERTa [24, 25], a feature concatenation operation is employed:

$$h_j^{\text{multi}} = \bigoplus_{k \in Y} H_j^{(L,k)}[0,:] \tag{5}$$

where Y denotes the set of classification labels. The probability output of each model is generated through a composite transformation:

$$p_i = \text{Softmax}\left(W_{c(i)} \cdot \text{GELU}\left(W_{h(i)} h_i + b_{h(i)}\right) + b_{c(i)}\right) \tag{6}$$

where $W_{h(i)} \in \mathbb{R}^{d_h \times d}$ and $W_{c(i)} \in \mathbb{R}^{|Y| \times d_h}$ form a two-layer projection structure. This mechanism ensures that the semantic information is effectively captured and transformed for subsequent fusion.

2.3 Hierarchical Decision Fusion

The hierarchical decision fusion mechanism [26] constitutes the core innovation of our framework. Model performance is evaluated using a hybrid metric:

$$s_i = \lambda \cdot F1_{i_{\text{micro}}} + (1-\lambda) \cdot \frac{1}{|Y|} \sum_{k \in Y} \text{Precision}_i(k) \tag{7}$$

where $\lambda \in [0,1]$ dynamically balances the micro F1 score and the average precision across categories. The decision process is bifurcated into two stages:

1)Majority Voting: If the category satisfies $V(k) = \sum_{i=1}^{5} \delta(\hat{y}_i, k) \geq 4$, the absolute majority vote result is directly adopted.

2)Weighted Fusion: Otherwise, the weighted fusion mechanism is activated, selecting the top 3 performing models. Stability weights are calculated as:

$$w_i = \frac{1}{1+\epsilon_i} \tag{8}$$

where $\epsilon_i = \frac{1}{T} \sum_{t=1}^{T} \delta(\hat{y}_i^{(t)}, \hat{y}_i)$ reflects model prediction consistency.

The final decision $\hat{y}_{\text{vote}}$ is determined by:

$$\hat{y}_{\text{vote}} = \arg\max_k \sum_{i \in I_{\text{top3}}} \left(\frac{w_i}{\sum_j w_j} \cdot p_i(k) \right) \tag{9}$$

This two-stage approach ensures robust decision-making by combining the strengths of majority voting and weighted fusion. The specific implementation steps of this two-stage decision fusion process are shown in Algorithm 1.

2.4 Multi-Granularity Feature Engineering

The multi-granularity feature engineering phase enhances classification signals through deep information fusion. The semantic embedding concatenation operation is defined as:

$$H_{\text{concat}} = \bigoplus_{i=1}^{3} h_i \oplus \bigoplus_{j=4}^{5} h_j^{\text{multi}} \tag{10}$$

forming a high-dimensional joint feature space. The decision signal encoding is:

$$o = [\mathbb{I}(\hat{y}_{\text{vote}} = E), \mathbb{I}(\hat{y}_{\text{vote}} = S), \mathbb{I}(\hat{y}_{\text{vote}} = G)] \tag{11}$$

which is concatenated with the semantic features to form the final feature vector:

$$X_{\text{final}} = H_{\text{concat}} \oplus o \tag{12}$$

This fusion strategy effectively combines semantic information with decision preferences, enhancing the model's ability to distinguish between different classes. The specific implementation details of this feature fusion process are shown in Algorithm 1.

2.5 Random Forest Classifier

The random forest classifier is employed to enhance generalization ability through ensemble learning. Each decision tree utilizes the information gain criterion:

$$IG(\nu) = H(\nu) - \sum_{c \in \{\text{left,right}\}} \frac{|D_t^c|}{|D_t|} H(\nu_c) \tag{13}$$

where the entropy calculation is:

$$H(\nu) = -\sum_{k \in Y} p_\nu(k) \log p_\nu(k) \tag{14}$$

ensuring optimal partitioning of the feature space. Forest-level prediction is achieved through the average voting mechanism:

$$\hat{y}_{\text{final}} = \arg\max_k \left(\frac{1}{T} \sum_{t=1}^{T} q_t(k) \right) \tag{15}$$

which effectively mitigates the overfitting risk associated with individual trees, thereby improving the model's robustness and accuracy. The training and prediction process of the random forest classifier is shown in Algorithm 1.

2.6 Theoretical Guarantees

Our proposed framework provides several theoretical guarantees that ensure its effectiveness and reliability:

- **Completeness**: The union of the support sets of all models covers the entire label space, i.e., $\bigcup_{i=1}^{n} \text{supp}(p_i) = Y$.
- **Consistency**: As the number of models increases, the probability of the predicted label matching the true label approaches 1 almost surely, i.e., $\lim_{n\to\infty} P(\hat{y} = y_{\text{true}}) = 1$ a.s.
- **Robustness**: The framework is robust to perturbations, with the norm of the gradient of the predicted label with respect to perturbations bounded by $\|\frac{\partial \hat{y}}{\partial \epsilon}\|_2 \leq C\sqrt{d}$.

2.7 Implementation Details

The implementation of the proposed framework meticulously follows the outlined theoretical framework and algorithmic steps (Algorithm 1). Each stage is designed to optimize performance while ensuring computational efficiency. The text preprocessing stage dynamically adjusts tokenization based on model-specific requirements. The semantic embedding extraction phase leverages multi-level encoding to capture rich semantic information. The hierarchical decision fusion mechanism combines majority voting and weighted fusion to ensure robust decision-making. Finally, the random forest classifier aggregates predictions to produce the final output, ensuring high accuracy and generalization ability.

3 Experiment

3.1 Data Preparation

3.1.1 Data Collection

The construction process of the corpus includes three phases: data collection, data preprocessing, and quality assessment. To ensure the industry representativeness and data quality of the samples, the selection criteria follow a dual dimension: according to the Bloomberg industry classification standard, select the top 20% listed companies by market capitalization in each sub-sector, and simultaneously combine the Refinitiv ESG rating system to screen out high-quality companies with an AAA rating, thereby constructing a benchmark corpus that encompasses both industry breadth and depth of ESG practice.

Algorithm 1: En-BERT Annotation Model

Input: Raw text T, Pre-trained models $M = \{M_1, M_2, \ldots, M_n\}$
Output: Final prediction $\hat{y}_{\text{final}}$

Preprocess text T to generate token sequences T_{proc} for each model M_i;
for *each model* M_i **do**
    Extract semantic embeddings H_i using the model's tokenizer and encoder;
    Compute probability output p_i using the composite transformation;
Evaluate model performance s_i using the hybrid metric;
Perform hierarchical decision fusion: **if** *majority vote condition is met* **then**
    $\hat{y}_{\text{vote}} \leftarrow$ majority vote result;
else
    Select top 3 models based on s_i;
    Compute stability weights w_i for the top 3 models;
    Calculate weighted probabilities $S(k)$ for each category k;
    $\hat{y}_{\text{vote}} \leftarrow \arg\max(S(k))$;
Concatenate semantic embeddings and decision signals to form X_{final};
Train random forest classifier on X_{final};
$\hat{y}_{\text{final}} \leftarrow$ random forest prediction;
return $\hat{y}_{\text{final}}$;

This study defines ESG-related text as an independent semantic unit containing at least one core ESG element (environmental, social, or governance dimension).We use manual screening to collect English news texts from two authoritative sources from 2024 to the present:

1) Direct Disclosure Channels: The ESG section of corporate websites and annual sustainability reports;

2) Indirect Verification Channels: Including professional ESG media (ESG Today, ESG News) and international mainstream financial media (Reuters, Bloomberg Businessweek), and cross-verify data from multiple sources to ensure the objectivity and completeness of the information sources.

As illustrated in Figure 3, a word cloud generated from the collected ESG-related texts is presented.



Figure 3: Word Cloud of Collected ESG-Related Texts

3.1.2 Data Preprocessing

To ensure the precision of text annotation, the annotation work combines manual and LLMs approaches. Among them, the LLMs uses ChatGPT for annotation, accounting for 20%.Manual annotation accounts for 80%. Two professional annotators who meet the annotation standards are selected through preliminary testing to participate in this task using a double-blind annotation method.The two annotators independently annotate the news texts.

The ESG classification system in this study adopts the three dimensions of a three-dimensional binary representation framework. We set up binary variables for the three ESG dimensions, i.e., marking their relevance with "0" and "1" based on whether the news text content involves the E, S, or G dimensions.

3.1.3 Quality Assessment

We use Cohen's Kappa Coefficient[27]to measure the consistency among annotators:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \tag{16}$$

Calculate the corresponding Kappa Coefficient scores for the ESG binary classifications.The κ values for the three dimensions are: 0.89 for the E dimension, 0.87 for the S dimension, and 0.84 for the G dimension, all reaching the "almost complete agreement" level [28].

3.2 Experiment Design

This study aims to treat various methods fairly, so conventional and effective models are selected to fairly demonstrate the advantages of this research, using conventional empirical parameters. Specifically, the traditional models selected are as follows: {Dictionary-based method: TF-IDF, Machine learning method: Logistic Regression, Ensemble learning: Random Forest, Neural network: Multilayer Perceptron}. The BERT models selected are {bert-base-cased, bert-base-uncased, DeBERTa, DistilBERT, RoBERTa}, all using the base version. The ensemble models include Vote-Bert and En-Bert.

Regarding the training part, the parameter space used in this study is shown in the Table 1 below. The traditional methods use TF-IDF for tokenization, and BERT uses the cross-entropy loss function with an Adam optimizer with weight decay. All experiments are conducted in a system environment of Ubuntu 22.04, using 8 NVIDIA 3090 GPUs and PyTorch.Regarding the evaluation part, the experiments use four metrics to assess model performance: Accuracy, Precision, Recall, and F1 score. Additionally, binning is used with bin=50, to obtain the mean and standard deviation to evaluate the robustness of different models.

Table 1: Parameters for models

TF-IDF					BERT			
	Dictionary Method	Machine Learning	Ensemble Learning	Neural Network		Standard BERT	Vote-BERT	En-BERT
n_keywords	50	50	50	50	batch_size	512	512	512
max_features	1000	1000	1000	1000	num_epochs	5	5	5
max_iter	\	1000	\	100	max_length	64	64	64
n_estimators	\	\	50	\	n_estimators	\	\	50
lr	\	\	\	2.00×10^{-5}	lr	2.00×10^{-5}	2.00×10^{-5}	2.00×10^{-5}

3.3 Experiment Analysis

As shown in Table 2, the proposed En-BERT model outperforms traditional dictionary-based methods, conventional machine learning models, and existing LLMs in classification tasks. In the E, S, and G tasks, En-BERT achieves average accuracy rates of 99.78%, 99.66%, and 99.65%, respectively, with performance stability far exceeding other models. Compared to the accuracy of the dictionary-based method of 65. 09%, En-BERT demonstrates an improvement of more than 34 % , while its F1 score reaches 99. 77%, much higher than the dictionary method's 69.87%. Compared to conventional machine learning models, En-BERT not only maintains high precision but also significantly improves recall. For example, in the environmental label classification task, En-BERT achieves a recall rate of 99.63%, while the MLP model only reaches 91.70%, proving its stronger robustness to class imbalance. Furthermore, compared to the 671B parameter Deepseek model, En-BERT achieves small improvements in all metrics while significantly reducing computational cost consumption. A further

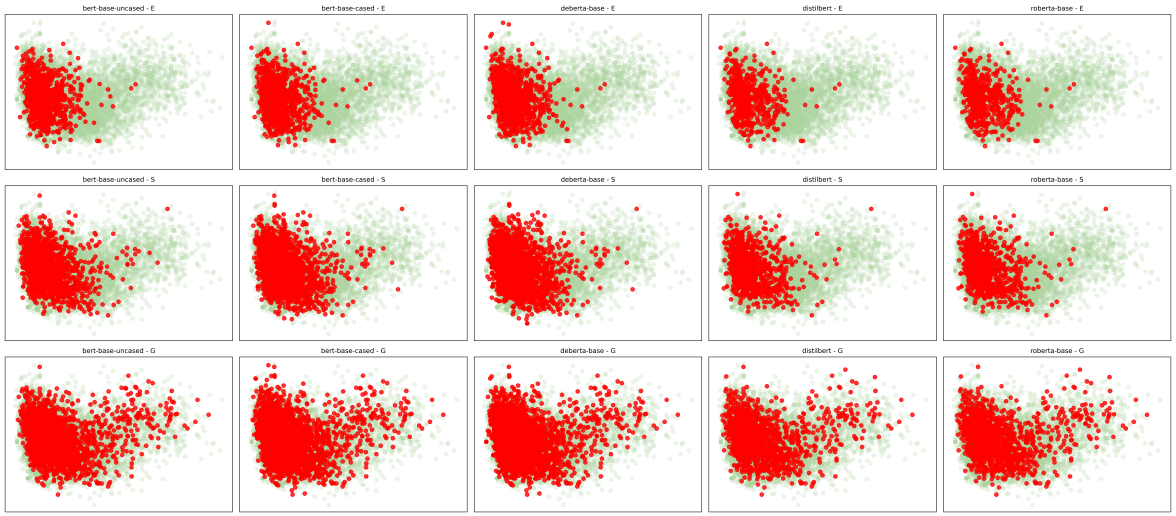


Figure 4: Visualization of ESG Dimension Predictions

comparison of En-BERT with BERT series variants and hard voting ensemble models shows that En-BERT demonstrates stable and significant performance advantages, as shown in Table 3 and Table 4. For instance, in the environmental label classification task, En-BERT's accuracy is 5.18 % higher than Bert-base-cased and 3.02 % higher than Roberta-base, while its F1 score also significantly outperforms Vote-BERT's 94.92%. As shown in Figure 4 traditional hard voting strategies tend to blur the classification boundaries due to the high consistency in predictions on misclassified samples among BERT variants, which reduces the robustness of the ensemble. In contrast, En-BERT mitigates the overfitting risk of a single deep learning model by introducing Random Forest as a heterogeneous base classifier. The strong ability of Random Forest to capture local textual features complements BERT's contextual semantic representation, thus inhibiting the accumulation of errors during the ensemble process. For example, in the social label classification task, En-BERT achieves a recall rate of 99.38%, while Vote-BERT only reaches 85.32%. Additionally, En-BERT adopts a dynamic weight allocation mechanism, which adaptively adjusts the voting weight between BERT and Random Forest based on validation set performance, further improving the model's ability to discriminate complex semantics. In the governance label classification task, En-BERT's precision reaches 99.73%, far surpassing the 71.14% to 84.97% of individual BERT models and variants, fully demonstrating the effectiveness of this method.

3.4 Ablation experiment

To evaluate the generalizability of the model, ablation studies were conducted on two additional ESG datasets provided by previous studies: the ESG-FTSE[1] and Nano-ESG datasets[2]. These datasets consist of UK-based and German news articles, respectively, which, in combination with the US-based dataset used in our primary experiments, collectively demonstrate the strong performance of En-BERT across data from different countries.

Moreover, the three data sets correspond to varying text lengths: short, medium, and long texts, enabling a comprehensive assessment of the model's capability under different textual conditions. Figure 4 presents the accuracy scores and error rates of various models across the two additional datasets.

[1]ESG-FTSE dataset comes from the dataset provided by the authors of the paper - *ESG-FTSE: A corpus of news articles with ESG relevance labels and use cases.*

[2]Nano-ESG datasetcomes from the dataset provided by public github in https://github.com/Bailefan/Nano-ESG from the paper - *Nano-ESG: Extracting Corporate Sustainability Information from News Articles.*

Table 2: En-BERT Classification results with tranditional models

Model	Dict	ML	EL	MLP	Deepseek	En-BERT
E_Accuracy	65.09 (±18.91)	92.65 (±16.65)	98.27 (±6.27)	93.93 (±11.93)	93.32 (±10.32)	99.78 (±3.78)
E_Precision	57.98 (±24.88)	95.28 (±27.43)	98.56 (±11.61)	95.01 (±18.09)	92.76 (±13.39)	99.92 (±4.92)
E_Recall	89.23 (±22.57)	88.55 (±18.98)	97.68 (±12.68)	91.70 (±25.03)	93.60 (±14.89)	99.63 (±8.72)
E_F1	69.87 (±19.87)	91.62 (±15.62)	98.08 (±8.61)	93.17 (±15.75)	93.18 (±12.34)	99.77 (±4.53)
S_Accuracy	53.45 (±23.45)	87.81 (±13.81)	97.14 (±11.14)	89.70 (±15.70)	92.10 (±13.10)	99.66 (±3.66)
S_Precision	48.01 (±33.73)	89.47 (±22.80)	98.14 (±13.93)	89.58 (±22.91)	91.51 (±20.25)	99.80 (±6.94)
S_Recall	80.55 (±30.55)	81.81 (±26.26)	95.23 (±22.15)	86.37 (±31.37)	91.91 (±17.09)	99.38 (±8.91)
S_F1	59.71 (±36.98)	85.14 (±18.47)	96.58 (±12.13)	87.72 (±22.86)	91.71 (±14.92)	99.58 (±5.46)
G_Accuracy	43.81 (±16.19)	86.21 (±16.21)	96.69 (±8.69)	88.50 (±14.50)	89.86 (±10.86)	99.65 (±3.65)
G_Precision	29.70 (±21.37)	84.29 (±50.96)	96.92 (±21.92)	80.89 (±60.89)	84.62 (±48.76)	99.73 (±9.73)
G_Recall	85.86 (±29.61)	57.26 (±40.60)	90.20 (±36.36)	72.08 (±59.58)	85.66 (±38.12)	98.96 (±17.15)
G_F1	43.67 (±29.04)	67.13 (±44.91)	93.15 (±23.15)	75.41 (±60.03)	85.14 (±41.73)	99.32 (±9.32)

4 Further Analysis

4.1 Impact of Events on ESG Ratings

In the evaluation of corporate performance, E, S, and G factors have emerged as essential indicators for measuring a firm's sustainability and long-term competitiveness. Our study aims to analyze the influence of specific events on corporate ESG ratings and to validate the effectiveness of an improved Ensemble BERT Random Forest model in enhancing the accuracy of ESG textual assessments.

4.1.1 Limitations of Conventional ESG Assessments

Traditional ESG rating systems quantify a firm's performance in environmental protection, social responsibility, and corporate governance to provide decision support for investors. However, existing evaluation methods exhibit the following limitations:

1) Reliance on single-source data, which makes it difficult to capture the dynamic changes within a firm;

2) Inconsistent evaluation criteria, resulting in a pronounced subjectivity in the ratings;

3) An inability to promptly capture significant events that affect a firm's ESG performance.

To address these issues, this study proposes an enhanced approach based on an Ensemble BERT Random Forest model. The method leverages machine learning to integrate multi-source data and dynamic evaluation metrics, thereby improving both the precision and timeliness of ESG ratings.

4.1.2 Case Study: Changes in Apple Inc.'s ESG Ratings

Taking Apple Inc. as an illustrative example, our study investigates the fluctuations in its ESG ratings from March 2024 to March 2025, with a focus on examining the impact of major events on ratings across different dimensions. Figure 5 presents the predictive outcomes of three distinct models:

- The Final model, which corresponds to the Ensemble BERT Random Forest model;
- The Simple BERT model;

Table 3: En-BERT Classification results with various BERT models

Model	BERT1	BERT2	BERT3	BERT4	BERT5	Vote-BERT	En-BERT
E_Accuracy	94.60 (±12.60)	95.00 (±9.00)	93.32 (±11.32)	96.76 (±8.76)	96.76 (±8.76)	95.47 (±11.47)	99.78 (±3.78)
E_Precision	97.30 (±18.35)	95.94 (±16.78)	94.43 (±15.27)	96.68 (±14.07)	96.68 (±14.86)	96.18 (±14.93)	99.92 (±4.92)
E_Recall	90.81 (±19.98)	93.22 (±18.22)	90.99 (±24.32)	96.28 (±14.46)	96.30 (±12.97)	93.90 (±22.48)	99.63 (±8.72)
E_F1	93.81 (±14.74)	94.44 (±11.68)	92.53 (±15.11)	96.41 (±10.69)	96.42 (±9.24)	94.92 (±13.84)	99.77 (±4.53)
S_Accuracy	84.53 (±14.53)	89.96 (±13.96)	83.32 (±15.32)	94.48 (±10.48)	94.48 (±10.48)	91.05 (±13.05)	99.66 (±3.66)
S_Precision	86.11 (±32.26)	91.83 (±23.41)	84.79 (±30.25)	95.68 (±17.90)	95.54 (±23.31)	93.43 (±20.70)	99.80 (±6.94)
S_Recall	77.11 (±27.11)	84.55 (±26.66)	75.27 (±35.27)	91.61 (±21.02)	91.59 (±21.00)	85.32 (±26.49)	99.38 (±8.91)
S_F1	80.97 (±21.51)	87.78 (±19.36)	79.31 (±33.16)	93.45 (±18.45)	93.38 (±13.38)	88.96 (±20.39)	99.58 (±5.46)
G_Accuracy	82.47 (±18.47)	87.40 (±13.40)	81.57 (±17.57)	90.65 (±14.65)	90.66 (±10.66)	88.26 (±14.26)	99.65 (±3.65)
G_Precision	71.14 (±58.64)	80.90 (±43.40)	72.98 (±47.98)	84.97 (±34.97)	84.77 (±30.93)	84.22 (±39.78)	99.73 (±9.73)
G_Recall	55.05 (±44.95)	67.35 (±38.77)	46.67 (±40.00)	77.96 (±47.19)	77.99 (±35.13)	67.36 (±34.03)	98.96 (±17.15)
G_F1	61.00 (±48.50)	72.46 (±34.96)	55.66 (±41.37)	80.60 (±40.60)	80.56 (±30.56)	74.01 (±31.16)	99.32 (±9.32)

Note: The BERT Number in the table is an abbreviation, and BERT1 to BERT5 indicate the model bert-base-cased, bert-base-uncased, deberta-base, distilbert, roberta-base

- The Ensemble BERT model.

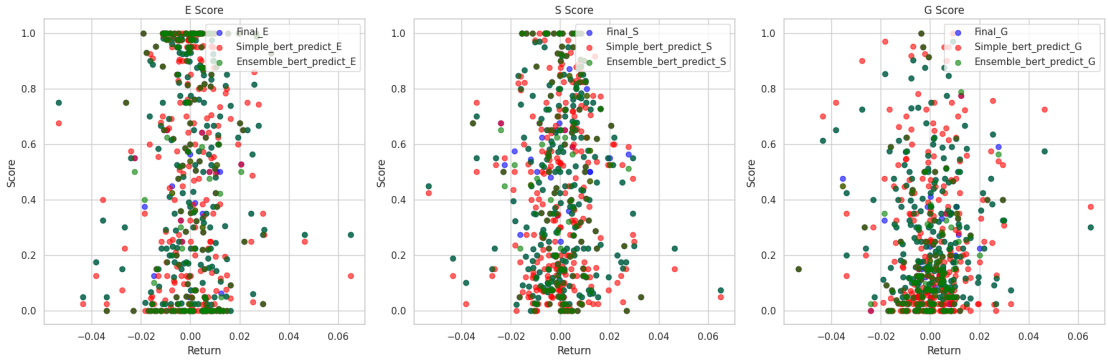


Figure 5: Scatter Plot Depicting the Relationship between ESG Scores and Stock Returns across Different Dimensions

Comparative analysis reveals that the Ensemble BERT model exhibits higher accuracy and stability.

In the Figure 6, E, S, and G denote the ESG ratings corresponding to the Environmental, Social, and Governance dimensions, respectively. Under the influence of specific events, Apple Inc.'s ESG ratings exhibited significant fluctuations at various time points. For example:

- In May 2024, Apple Inc. announced initiatives to reduce greenhouse gas emissions, markedly enhancing its environmental rating;
- In November 2024, by endorsing responsible AI guidelines, the company achieved an improved governance rating;

Table 4: Further experiments were conducted on ESG-FTSE and Nano-ESG datasets

Model	ESG-FTSE			Nano-ESG		
	E_Acc	**S_Acc**	**G_Acc**	**E_Acc**	**S_Acc**	**G_Acc**
Dict	12.91 (±11.09)	8.65 (±11.35)	15.52 (±28.92)	55.67 (±24.33)	32.61 (±22.61)	27.10 (±22.90)
ML	93.67 (±9.67)	95.01 (±7.01)	88.73 (±12.73)	92.26 (±14.26)	91.14 (±17.14)	91.19 (±13.19)
EL	98.71 (±4.71)	98.91 (±21.13)	97.44 (±9.44)	97.98 (±13.98)	97.57 (±9.57)	97.44 (±7.44)
MLP	96.37 (±7.48)	95.37 (±7.37)	92.97 (±10.97)	96.17 (±14.17)	96.06 (±10.06)	96.31 (±10.31)
Deepseek	77.77 (±22.23)	50.62 (±25.62)	62.34 (±16.34)	82.14 (±18.64)	71.98 (±22.14)	70.34 (±12.34)
BERT1	89.44 (±15.44)	94.37 (±8.37)	86.05 (±16.05)	96.54 (±10.54)	96.35 (±10.35)	96.29 (±10.29)
BERT2	89.21 (±11.43)	94.37 (±10.37)	86.28 (±18.28)	96.50 (±10.50)	96.15 (±14.15)	95.94 (±13.94)
BERT3	89.33 (±9.33)	94.48 (±8.48)	86.05 (±30.49)	96.59 (±8.59)	96.24 (±12.24)	96.12 (±12.12)
BERT4	91.54 (±9.54)	94.46 (±6.46)	86.95 (±20.28)	95.65 (±13.65)	94.60 (±12.60)	94.17 (±12.17)
BERT5	91.43 (±9.43)	94.46 (±14.46)	87.18 (±17.18)	95.65 (±13.65)	94.60 (±14.60)	94.17 (±12.17)
Vote-BERT	89.44 (±11.44)	94.48 (±6.48)	86.23 (±11.77)	96.65 (±10.65)	96.24 (±12.24)	95.98 (±11.98)
En-BERT	99.95 (±1.95)	99.97 (±1.97)	100.00 (±0.00)	99.98 (±1.98)	99.99 (±1.99)	99.98 (±1.98)

- In January 2025, due to litigation in the Democratic Republic of Congo concerning conflict minerals, its social rating experienced a decline.

4.1.3 Analysis of Data Characteristics and Model Validation

The analysis indicates that the Ensemble BERT model is capable of effectively capturing variations across the Environmental, Social, and Governance dimensions and dynamically adjusting ratings in response to significant events. Compared with conventional models, this improved ESG evaluation approach exhibits higher precision and can promptly reflect a firm's current performance, thereby offering investors and other stakeholders a more scientifically sound and reliable basis for decision-making.

The findings further demonstrate that the enhanced Ensemble BERT model not only improves the predictive performance of ESG ratings but also underscores the importance of ESG assessments in the analysis of corporate sustainability.

4.2 Visualization Analysis of the Relationship between ESG Scores and Stock Returns

To further illustrate the differences in the explanatory power of various ESG measurement methods with respect to stock returns, our paper presents scatter plots of ESG scores across the three dimensions (E/S/G) versus individual stock returns, as shown in Figure 5. In the figure, different colored points correspond to ESG scores derived from distinct sources:

- Blue points represent conventional ESG ratings (Final);
- Red points denote outputs from the simple BERT model (Simple_Bert_predict);
- Green points correspond to the Ensemble BERT + Random Forest model (Ensemble_Bert_predict), i.e., the improved model proposed in this study.

The figure yields the following key observations:

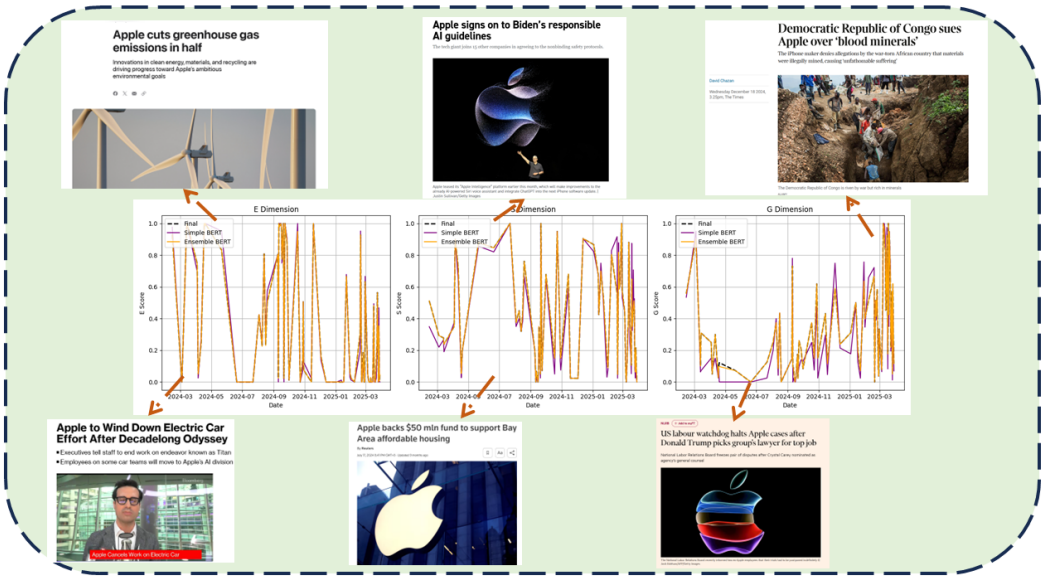


Figure 6: ESG Major Event Map of Apple Company

1) Wider Coverage of Stock Returns: The green points display a noticeably more dispersed distribution along the horizontal axis (Return) compared to the red and blue points, especially covering a broader range of extreme return values. This indicates that the integrated model has enhanced capability in identifying high-volatility stocks.

2) Enhanced Rating Differentiation: Along the vertical axis (ESG Score), the green points exhibit a more polarized distribution, particularly in the Environmental dimension, demonstrating that the model possesses a significantly higher discriminatory power in evaluating ESG performance.

3) Clearer Point Cloud Structure: For the Social and Governance dimensions, the green points show moderate density and well-concentrated clustering, thereby avoiding the "grouping" phenomenon observed in other models. This reflects improvements in the model's semantic understanding and output stability.

4) Better Alignment with Regression Trends: Although an explicit regression curve is not drawn, the distribution of points suggests that the green points are closer to the central trend line across all dimensions, signifying a closer linear relationship between the model's outputs and actual returns.

4.3 Empirical Analysis

To evaluate the effectiveness of the improved ESG textual measurement model in explaining stock returns, this study performs comparative regression analyzes based on the output of the constructed BERT model and conventional ESG rating results. Specifically, different measurement methods across the three ESG dimensions are used as independent variables, with stock returns serving as the dependent variable. A series of Ordinary Least Squares (OLS) regression models are constructed, and robust standard errors—consistent with heteroscedasticity and autocorrelation (HAC)—are employed to mitigate potential heteroscedasticity and autocorrelation issues in the time-series data.

4.3.1 Model Configuration and Variable Description

The independent variables include the following three ESG measurement methods:

- **Simple_Bert_predict**: ESG textual ratings directly produced by the basic BERT model;
- **Ensemble_Bert_predict**: ESG textual ratings obtained by integrating the basic BERT outputs with a Random Forest approach, representing the improved model proposed in this study;

- **Final**: Conventional ESG ratings available in the data source, which serve as the baseline for comparison.

The dependent variable is the individual stock **return** corresponding to the respective time period, used to assess the explanatory power of the various ESG measurement methods with respect to market performance.

The general form of the regression model is given by:

$$Return_t = \alpha + \beta \cdot ESG_{i,t} + \varepsilon_t \tag{17}$$

where $ESG_{i,t}$ denotes the ESG rating under a specific method, $Return_t$ represents the stock return for the period, and ε_t is the error term.

4.3.2 Analysis of Regression Results

Table 5 summarizes the regression R^2 values obtained from the Simple BERT and Ensemble BERT models across different ESG dimensions:

Table 5: Comparison of Regression R^2 Values across ESG Dimensions (Unit: %)

Dimension	Regression Method	Average R^2 Value
E	Simple_bert_predict_E	1.80
	Ensemble_bert_predict_E	**2.50**
S	Simple_bert_predict_S	1.00
	Ensemble_bert_predict_S	**1.20**
G	Simple_bert_predict_G	0.70
	Ensemble_bert_predict_G	**0.90**

As evident from the table, the **Ensemble_bert_predict model outperforms the basic BERT model across all dimensions**. This indicates that integrating the Random Forest method can effectively enhance the precision of ESG textual ratings and strengthen their ability to explain market performance. Although the overall R^2 values are relatively modest, this is consistent with the inherently weak economic rationale linking ESG information with short-term returns, thereby demonstrating that the model partially captures the connection between non-financial information in corporate texts and market reactions.

4.3.3 Result

The empirical results validate the effectiveness of the proposed **BERT + Random Forest ensemble model for ESG textual measurement**, with the most significant improvement observed in the Environmental dimension. This suggests that the incorporation of semantic analysis with nonlinear ensemble algorithms can enhance the identification of corporate performance in the construction of ESG rating systems. Future research may further incorporate cross-sectional control variables and industry-fixed effects to enhance the robustness and generalizability of the model.

5 Related Works

5.1 Evolution of ESG Text Annotation

Early ESG text annotation primarily relied on lexicon-based methods, where researchers constructed ESG-related word lists or keyword rules to identify sustainability-related content in corporate annual reports or CSR disclosures[29]. To enhance vocabulary-level analysis, various ESG-specific dictionaries were developed, often categorized by Environmental (E), Social (S), and Governance (G) dimensions, and widely applied in automated annotation tasks. One representative method is the unigram-based ESG dictionary proposed by Baier et al., which performs ESG content extraction via keyword matching[30]. However, such methods rely heavily on expert curation and manual scoring, leading to delayed updates, non-transparent standards, and high subjectivity. These limitations make

traditional lexicon-based approaches prone to semantic ambiguity, polysemy, and domain shifts, ultimately affecting the coverage and accuracy of annotations[31].

As the limitations of lexicon-based approaches became increasingly evident, researchers began exploring traditional machine learning methods for ESG text annotation. A typical approach is to transform text into feature vectors (e.g., TF-IDF matrices or sentiment word counts), then apply classifiers such as support vector machines (SVMs) or random forests for ESG category or sentiment labeling[32, 33]. Compared to rule-based methods, machine learning models can leverage data-driven learning to improve annotation accuracy. However, these models still rely on manually engineered features and struggle to capture deep semantic structures. Their performance remains suboptimal in complex corpora with high polysemy or industry-specific expressions.

Building on traditional machine learning, ensemble learning has been introduced as a strategy to enhance model stability and generalization in ESG annotation tasks[34]. Ensemble methods combine multiple base learners (such as decision trees, SVMs, or naive Bayes classifiers) at either the model level or the output level to mitigate the bias and variance issues of single models. Luo et al.[35] compared SVMs, logistic regression, and random forest models for detecting exaggerated or misleading expressions in ESG reports, and found that the Bagging-based random forest model performed best when dealing with redundant or semantically complex ESG text. Other studies have adopted voting or weighted fusion approaches to combine traditional classifiers for multi-label ESG risk classification in news content[34]. These ensemble techniques not only alleviate overfitting and sparsity issues in traditional models but also laid the groundwork for integrating deep learning and pretrained language models.

The adoption of deep learning brought a new phase to ESG text annotation. Models such as convolutional neural networks (CNNs) and long short-term memory networks (LSTMs) began to be applied to ESG-related text classification and sentiment analysis tasks[36]. Compared to conventional models, deep neural networks can automatically learn feature representations from large-scale corpora, reducing the need for manual feature design. Studies have shown that multi-layer perceptrons (MLPs), CNNs, and RNNs outperform traditional classifiers like naive Bayes and SVMs in text classification tasks[37]. CNNs can capture local textual patterns to detect ESG polarity, while LSTMs are effective in modeling context-dependent meanings. Although deep learning significantly improved ESG annotation effectiveness, it still faces limitations in consistency and robustness for practical applications.

LLMs are now transforming the technical paradigm of ESG text annotation. Compared with traditional methods, LLMs leverage large-scale pretraining to achieve stronger semantic understanding and contextual modeling capabilities, showing excellent performance in classification and information extraction tasks[38]. Domain-specific BERT variants such as FinBERT[39] have been fine-tuned on financial and sustainability texts, significantly improving the ability to identify environment, social, and governance content in ESG documents. Among them, models like FinBERT-ESG[40] support sentence-level ESG classification and have become early influential benchmark tools. These models outperform traditional methods in terms of accuracy and generalization, and better handle ambiguous ESG expressions and terminological diversity, gradually becoming a cornerstone of automated ESG annotation systems. However, LLMs still face significant challenges in real-world applications. On one hand, these models often contain billions of parameters, leading to long training and inference times and high computational resource consumption. Simply loading such models often requires high-performance GPUs, limiting deployment in small and medium-scale environments[41]. On the other hand, due to the lack of fine-grained supervision, LLMs are prone to hallucination in structured annotation tasks, resulting in inaccurate or inconsistent outputs[42]. Furthermore, despite expanded context windows, current LLMs still struggle to maintain global coherence across long and multi-paragraph texts. In ESG applications, this leads to difficulties in entity boundary control and semantic consistency. In contrast, BERT and its variants offer a simpler architecture, moderate parameter size, mature fine-tuning mechanisms, and high inference efficiency, making them the preferred choice for ESG annotation tasks that demand a balance of performance and cost-effectiveness.

5.2 Limitations of Current BERT Baselines

BERT models are typically used in text annotation tasks through an "encoder + classification head" architecture, where the input is encoded via a bidirectional Transformer and then passed to a softmax or CRF layer for token-level or sequence-level labeling[43]. As a representative pretrained language model, BERT achieved breakthrough performance across many NLP tasks[44, 45], yet as a single

model, it still faces multiple limitations in ESG text annotation. Empirical studies have shown that in sentence-level ESG labeling tasks, single BERT models often suffer from label bias. For example, in FinBERT-ESG experiments, the model tended to classify neutral sentences as "Governance" due to the frequent appearance of governance-related language in corporate reports[46]. Furthermore, single BERT models often exhibit instability across different training and test sets, showing high sensitivity to hyperparameter configurations. These issues are especially prominent in ESG tasks, which are typically low-resource and highly context-dependent, leading to poor generalization across companies and industries[47].

To address these problems, researchers have proposed various structural improvements to BERT. One direction is RoBERTa, which improves the robustness of the model and the representational power by increasing the size of the pre-training corpus, extending the training time and removing the prediction objective of the next sentence[48]. Another direction is BART, which combines BERT's encoder with GPT-style decoders and is widely used in text summarization, question answering, and generative annotation tasks[49]. Notably, recent studies show that BERT-based models generally outperform LLMs in classification tasks, especially under conditions of limited labels or strict precision requirements, while LLMs demonstrate stronger robustness to textual perturbations and ambiguity. Moreover, compared to weakly supervised methods, AI-generated labels with human supervision have shown better classification outcomes, emphasizing the importance of integrating AI annotations with human oversight[50]. This insight is particularly valuable for ESG tasks, which require high-quality annotation standards. Despite these structural enhancements, BERT still faces issues of inconsistent output and label bias in ESG applications. To tackle this, this study integrates multiple pretrained BERT models and proposes an ensemble BERT architecture that combines multiple outputs, aiming to provide a more practical and accurate solution for ESG text analysis.

5.3 Application of Ensemble Learning in Text Annotation

Ensemble learning, a classical strategy to improve model stability and predictive accuracy, has gained increasing attention in natural language processing, especially in text annotation tasks. Its core idea is to combine outputs from multiple base models to compensate for the limitations of a single model in terms of generalization, robustness, or local errors. Common ensemble techniques include Bagging (e.g., Random Forest), Boosting (e.g., AdaBoost, XGBoost), Voting, and Stacking. Several studies have successfully applied these methods to ESG-related tasks[51, 52]. For instance, Luo et al. used a Random Forest model to classify the sentiment of corporate ESG disclosures, and found that it was more robust than traditional SVMs in sparse feature scenarios[35].

In named entity recognition and sentiment classification tasks, ensemble methods have also proven effective. For example, Lee et al. combined BERT and ALBERT in a voting-based ensemble model for ESG text classification, achieving an accuracy of 80.79% with a batch size of 20[53]. Ensemble learning shows clear advantages in ESG annotation. On one hand, different BERT variants (such as FinBERT and RoBERTa) perform variably across different domain subsets, and integrating them enables information complementarity[54]. On the other hand, ESG annotation involves significant subjectivity, and predictions from a single model may deviate from human judgment. Aggregating predictions from multiple models can improve the interpretability and consistency of the results[55]. Therefore, in light of the limitations of current BERT baselines, introducing ensemble strategies is not only a reliable way to enhance annotation quality, but also provides a methodological foundation for building more trustworthy and generalizable ESG information extraction systems. This paper will further validate the effectiveness of the proposed ensemble BERT method through empirical comparison.

6 Conclusion

This paper introduces an innovative two-stage bias mitigation approach, BMIL, effectively addressing age-related biases in Large Language Models (LLMs) through a combination of Reinforcement Learning (RL), human-in-the-loop mechanisms and empathy ability of LLM. Our approach has demonstrated significant improvements in model fairness without the need to alter underlying model parameters, directly applying the strategies outlined in our introduction.

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